

Name Generator Documentation (v0.2.1)

The Name Source Generator is a powerful C utility designed to create culturally and historically authentic Finnish names. It uses decade-specific data to ensure generated names reflect the chosen time period (1860s–1920s), making it ideal for **Game Developers** and **Narrative Designers**.

1. Command-Line Usage

The program can be run with various flags to automate generation (namegen.exe [flags]) or in interactive mode (namegen.exe).

Flag	Full Name	Description	Example
-p <num>	--period <num>	Selects the historical time period (decade) for name generation. (1–7)	-p 4 (Selects Period 4)
-n <count>	--count <count>	Specifies the number of names or couples to generate.	-n 10
-g <M/F>	--gender <M/F>	Selects the gender of the generated names (M or F). Ignored if -P is used.	-g female
-P	--couple	Couple Mode: Generates a pair of names (Male + Female) that share the same generated surname. Forces surname generation.	-P
-S <name>	--set-last-name <name>	Sets a specific, fixed surname for all generated names. Overrides random surname generation.	-S Lehtonen
-L	--no-last-name	Excludes the surname from the generated output. Ignored if -S or -P is set.	-L
-m <chance>	--middle-chance <chance>	Sets the probability (0–100) of a middle name being included.	-m 75 (75% chance)
-M <max>	--max-middle-names <max>	Sets the maximum number of middle names to generate (1–3).	-M 2
-v	--verbose	Activates verbose/debug mode , showing data loading details and generation steps.	-v
-V	--version	Displays the program version number and exits.	-V
-h	--help	Displays the help documentation.	-h

2. Interactive Mode

If the program is run without any command-line flags, it enters **Interactive Mode**, guiding the user through the settings:

1. **Period Selection:** Displays available time periods and prompts for a choice (1-7).
2. **Couple Mode Toggle:** Prompts: **Generate names for a couple (M+F)? (1=Yes, 0=No, default 0):**
3. **Count:** Prompts for the number of names or couples to generate.
4. **Middle Name Chance & Max Count:** Prompts for probability and maximum middle names (1-3).
5. **Last Name Generation:** Asks whether to generate a surname (if **Couple Mode** is not enabled).

3. Data Integrity & Validation (v0.2.0)

The generator employs strict internal checks to ensure output quality, a critical feature for production environments.

- **Strict Name Validation:** All name files are filtered during the loading process. Only strings composed of **alphabetic characters**, spaces, and hyphens (-) are accepted as valid name components.
- **Error Prevention:** This validation automatically rejects data artifacts (e.g., "1870-29" or internal notes) from being included in the generated names.
- **Warning System:** When the verbose flag (-v) is used, the program prints a WARNING to stderr whenever an invalid name candidate is found and skipped during file loading.

4. Generation Logic Overview

- **Single Name:** Uses the first name and middle name list corresponding to the selected **Period** and **Gender**. Surname is selected from a separate Finnish-last-names.csv list (or overridden by -S).
- **Couple Generation (-P):**
 1. A random surname is chosen from the last name list (or overridden).
 2. The **Male** name is generated using the Male lists and the chosen surname.
 3. The **Female** name is generated using the Female lists and is assigned the *exact same surname* as the male name.
- **Middle Name Inclusion:** A middle name is added only if a random number check passes the configured **Middle Name Chance** (-m). This check repeats until the probability fails or the **Max Middle Names** (-M) limit is reached.



V. TODO / Next Steps

Goal	Description	Status
Output Control	Ability to select how many names are generated at once (e.g., 1 or 10) using a new flag (-n or --count).	done
Setting Flexibility	Allow the middle name probability to be adjusted (e.g., -m 80 for 80\% probability).	done
Genesim Integration	Preparation for use by the genesim project (library module).	Planned